

**A REPORT ON
WIDE AREA NETWORK (WAN) CONNECTION
USING CISCO PACKET TRACER**

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Subject: Networking / Computer Networks

Topic: WAN Configuration and Connectivity

Date of Submission:
24/09/2026

WAN Connection using Cisco Packet Tracer

Introduction

A Wide Area Network (WAN) is a network that connects computers and networks over a large geographical area. Unlike LAN (Local Area Network), which is limited to a small area like a home or office, WAN connects different cities or even countries.

In this experiment, we used **Cisco Packet Tracer** to understand how WAN connections work by connecting multiple routers using serial links.

Objective

- To understand how WAN connections are established
 - To configure routers for WAN communication
 - To verify connectivity between different networks
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Requirements

- Cisco Packet Tracer
 - 2 or more Routers (e.g., 2911)
 - Serial cables
 - PCs for testing connectivity
-

Network Topology

In this setup:

- Two routers are connected using a **serial cable (WAN link)**
- Each router is connected to a LAN with PCs
- WAN acts as a bridge between two different LANs

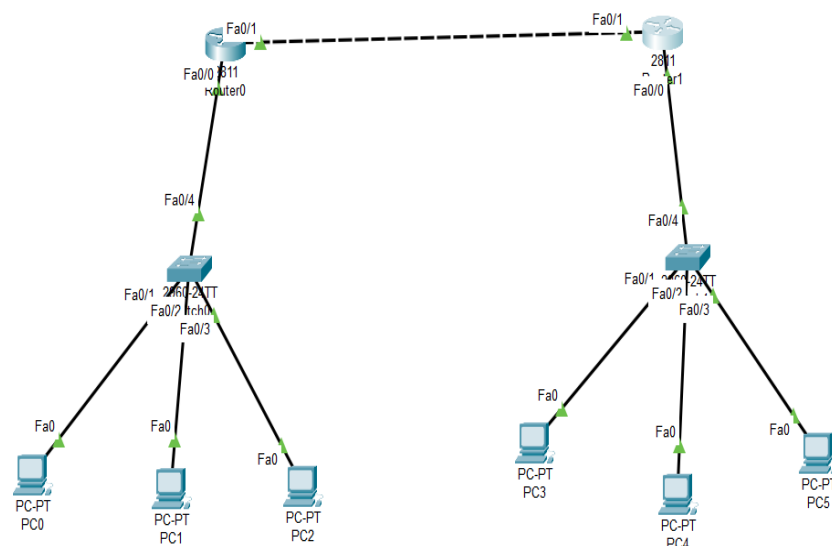
Steps to Configure WAN Connection

Step 1: Add Devices

- Place 2 routers in the workspace
- Add 2 Switch
- Add PCs and connect them to switch and then router using Ethernet cables

Step 2: Connect Routers (WAN Link)

- Use Copper Cable
- Connect:
 - Router1 → fa0/1
 - Router2 → fa0/1



Step 3: Configure IP Address on Router Interfaces

Router 0:

enable

configure terminal

interface fa 0/1

ip address 10.0.0.1 255.0.0.0

no shutdown

interface fa 0/0

ip address 192.168.1.45 255.255.255.0

no shutdown

Physical **Config** CLI Attributes

GLOBAL	FastEthernet0/0	
Settings		
Algorithm Settings		
ROUTING		
Static		
RIP		
SWITCHING		
VLAN Database		
INTERFACE		
FastEthernet0/0		
FastEthernet0/1		

Port Status	☒ On
Bandwidth	☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	0006.2AA1.4901
IP Configuration	
IPv4 Address	192.168.1.45
Subnet Mask	255.255.255.0
Tx Ring Limit	10

Router 1:

enable

configure terminal

interface fa0/1

ip address 10.0.0.2 255.0.0.0

no shutdown

interface fa0/0

ip address 172.16.0.45 255.255.0.0

no shutdown

physical **Config** CLI Attributes

GLOBAL	FastEthernet0/0	
Settings	Port Status ☒ On	
Algorithm Settings	Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto	
ROUTING	Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto	
Static	MAC Address 0001.C7AD.C901	
RIP	IP Configuration	
SWITCHING	IPv4 Address 172.16.0.45	
VLAN Database	Subnet Mask 255.255.0.0	
INTERFACE	Tx Ring Limit 10	
FastEthernet0/0		
FastEthernet0/1		

Step 4: Configure Routing

You can use **Static Routing**:

Router 1:

```
ip route 172.16.0.0 255.255.255.0 10.0.0.2
```

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

Static Routes

Network172.16.0.0

Mask255.255.0.0

Next Hop10.0.0.2

Add

Network Address

172.16.0.0/16 via 10.0.0.2

Remove

Equivalent IOS Commands

```
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#no ip route 172.16.0.0 255.255.0.0 10.0.0.2
Router(config)#ip route 172.16.0.0 255.255.0.0 10.0.0.2
Router(config)#
```

Router 2:

ip route 192.168.1.0 255.255.255.0 10.0.0.1

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

Static Routes

Network192.168.1.0

Mask255.255.255.0

Next Hop10.0.0.1

Add

Network Address

192.168.1.0/24 via 10.0.0.1

Remove

Equivalent IOS Commands

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#no ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#

Step 5: Configure PCs

- PC1:
 - IP: 192.168.1.2
 - Gateway: 192.168.1.45

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.45

DNS Server: 8.8.8.8

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::20C:CFFF:FEEB:E774

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

- PC3:
 - IP: 172.16.0.1
 - Gateway: 172.16.0.45

The screenshot shows the 'IP Configuration' window for the 'FastEthernet0' interface. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Desktop' tab is selected. The 'IP Configuration' section is active, showing the 'DHCP' radio button selected. The 'IPv4 Address' is set to '172.16.0.1', 'Subnet Mask' to '255.255.0.0', 'Default Gateway' to '172.16.0.45', and 'DNS Server' to '8.8.8.8'. The 'IPv6 Configuration' section shows the 'Static' radio button selected, with 'IPv6 Address' and 'Link Local Address' fields. The 'Link Local Address' is set to 'FE80::20C:85FF:FEB5:A640'. The '802.1X' section shows the 'Use 802.1X Security' checkbox unchecked, with 'Authentication' set to 'MD5' and 'Username' and 'Password' fields.

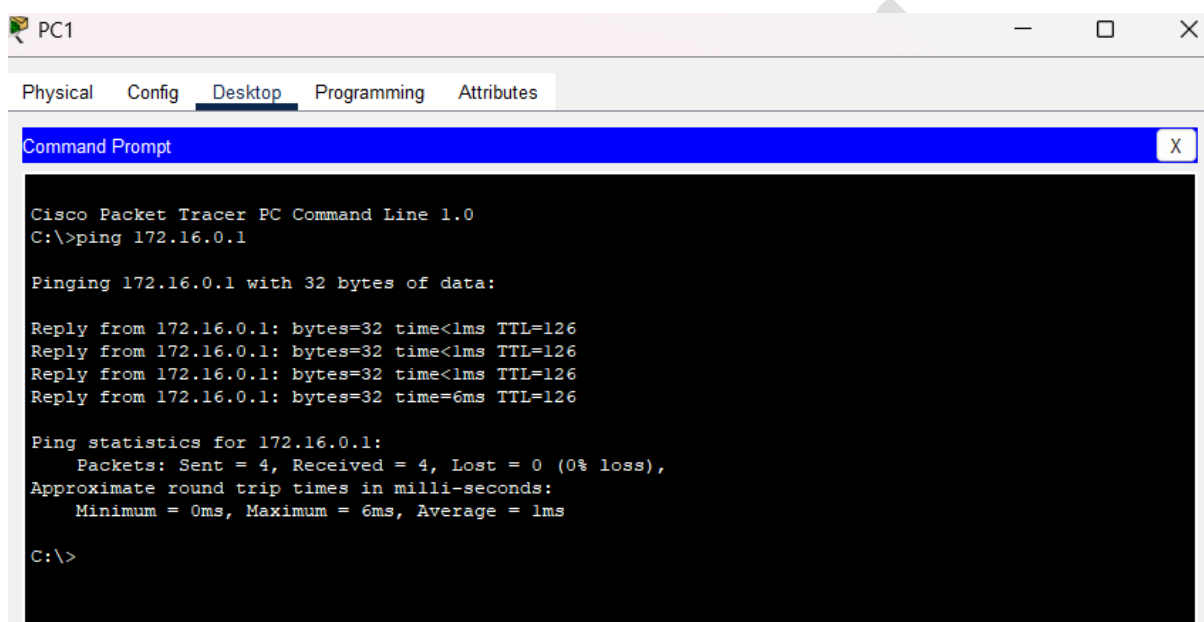
Interface	FastEthernet0
IP Configuration	
☒ DHCP	☐ Static
IPv4 Address	172.16.0.1
Subnet Mask	255.255.0.0
Default Gateway	172.16.0.45
DNS Server	8.8.8.8
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::20C:85FF:FEB5:A640
Default Gateway	
DNS Server	
802.1X	
☐ Use 802.1X Security	
Authentication	MD5
Username	
Password	

Step 6: Verify Connection

- Use ping command from PC1 to PC3
- If configuration is correct, you will receive reply messages

Example:

ping 172.16.0.1



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.0.1

Pinging 172.16.0.1 with 32 bytes of data:

Reply from 172.16.0.1: bytes=32 time<1ms TTL=126
Reply from 172.16.0.1: bytes=32 time<1ms TTL=126
Reply from 172.16.0.1: bytes=32 time<1ms TTL=126
Reply from 172.16.0.1: bytes=32 time=6ms TTL=126

Ping statistics for 172.16.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 1ms

C:\>
```

Result

The WAN connection was successfully established between two networks. Devices from different LANs were able to communicate with each other using routers connected via a serial WAN link.

Conclusion

This experiment helped in understanding how WAN connections work in real-world networking. We learned how to configure routers, assign IP addresses, and use routing to enable communication between different networks.